Group 07

WORKERS ASSIGNMENT FOR SELECTED ASSEMBLY LINE

GROUP MEMBERS

- S19809 - Kasun Dhanushka
- S19832 - Malith Mewanga
- S19836 - Thilina Sasanka
- S19845 - Hiruni Rathnaweera
- S19847 - Chamuditha Rowel
- S19850 - Thilini Sadarenu
- S19869 - Dineth Ashen

CONTENT

COMPANY OVERVIEW

Ctd…

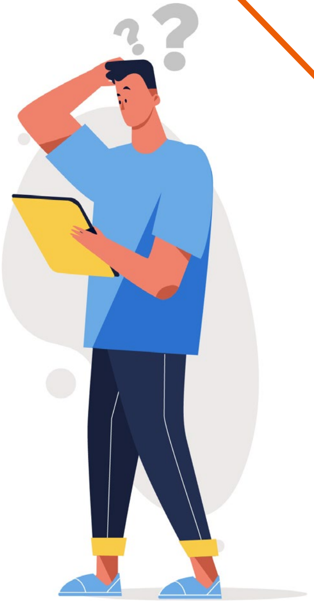
- Name : Emjay international (PVT) Ltd
- Products : Loungewear, underwear, active wear, and thermal wear
- Location : Panvila
- Main branch: Colombo
- Workers : 1200 employees

DEPARTMENTS

- Storing department
- Inspection area
- Cutting section
- Quality assurance department (overall procedure)
- Sewing department
- Finished goods warehouse

PROBLEMS IDENTIFIED

- Storage problems.
- Defects generating in sewing.
- Machine Downtime.
- Cover the targets before deadlines.



WHY WE SELECTED WORKERS ASSIGNING

Assigning Workers to Sewing Machines for maximizing skill score taken to finish a product in the given specific line.

DATA COLLECTION

- Collected skill scores for each worker (0-2 scale).
- Noted worker availability.
- Mapped tasks to required skill levels.
- Skill variation across workers.

DATA SET

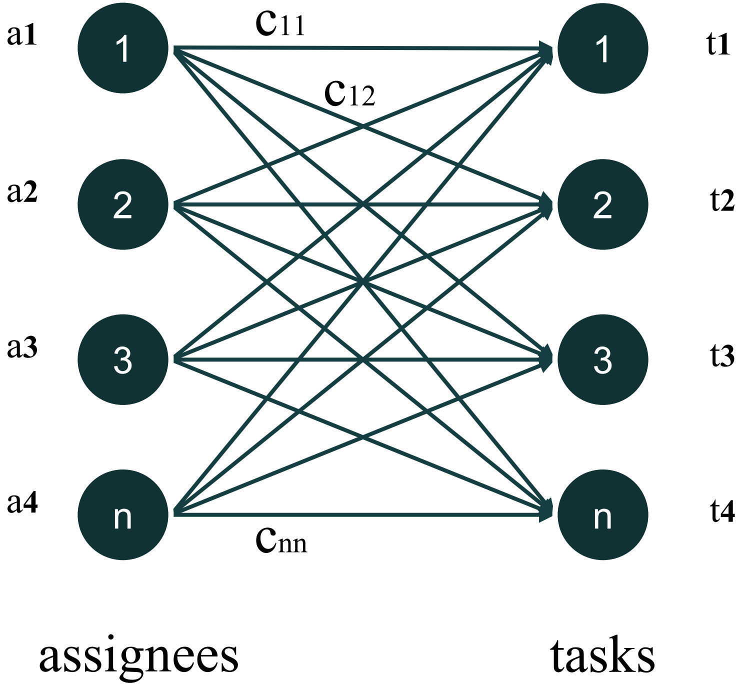
ID number	Front pieces darts	binding fold over elastic	sewing front pieces	sewing front pieces	Finishing front pieces & Thread cutting	Sewing back pieces(1)	cutting elastic band	Finishing pouch	Attach pouch	sewing pouch	Finishing back piece & Thread cutting	Sewing back pieces(2)	closing elastic	Sewing elastic	Sewing hem	ironing	packing(1)	packing(2)
E1118	1.19	0.95	1.85	1.85	0	0	1.83	1.76	1.19	1.23	0	1.57	1.63	1.28	0.86	0.89	1.56	0.63
E1246	0	1.84	0	0	1.16	0.77	1.19	0	0.76	1.4	1.59	1.09	0.68	1.15	1.09	1.95	1.68	1.75
E1156	1.58	1.68	0	0	0.8	1.69	0	0.74	0.81	1.6	1.38	1.43	0	1.14	1.72	1.18	1.95	0
E1143	1.59	0	0	0	1.49	1.58	0.85	0.74	0	0	0	0.64	1.06	1.01	0	0	1.42	1.55
E1179	0	1.86	1.18	1.18	0.74	1.51	0.73	0	1.58	0	0.55	0	0.73	0.83	1.89	0.74	0.81	0.46
E1099	0	1.08	0	0	1.87	1.46	1.42	1.43	0.74	1.11	1.37	1.55	0.68	0	1.51	1.02	1.62	0
E1233	0.69	0	0.63	0.63	0.77	0.44	0.7	0.99	0	1.85	0.62	0	0.53	0	0	1.27	1.69	1.38
E0999	0.94	1.56	1.38	1.38	0	1.83	0.51	1.85	1.95	1.81	1.89	1.94	1.72	1.29	1.75	0	1.16	1.27
E1567	0.48	1.55	0.3	0.3	1.36	0.33	1.12	0	0	0	1.09	1.58	1.94	0	1.94	1.92	0.69	1.24
E1589	1.22	1.86	1.28	1.28	1.28	0	1.53	0	1.27	1.24	0.81	1.92	1.12	1.29	1.65	0	0.3	0
E2004	1.74	0.85	0	0	1.32	0.89	0	0.94	0.92	1.19	0.75	1.84	0.53	0	0.84	1.49	0	0
E2014	0	0.84	1.8	1.8	0	0.85	0.77	1.79	1.38	0.9	1.72	0.91	0.11	1.18	1.07	1.84	1	1.94
E1887	1.91	1.68	1.77	1.77	1.89	0	1.31	0.6	1.73	0.59	0	0.7	1.14	0.63	1.08	0	1.71	0
E1934	0.9	0	0.56	0.56	1.31	0.93	1.81	0	1.57	0.43	0.4	1.42	0.97	1.5	0.72	1.88	0.59	1.67
E1857	0.97	0.75	0.97	0.97	0.75	0.55	0.99	0	0	0	0.43	0.16	1.51	0	1.76	0	0.72	0.82
E1469	0	0	0	0	0	1.76	1.71	1.72	1.53	1.02	1.91	1.37	1.07	0.69	0.64	1.02	0.67	1.05
E2069	1.6	1	0	0	0	1.08	1.94	0	1.6	0	1.89	0	1.16	1.57	0	1.32	0	0.63
E2071	1.94	0	1.46	1.46	0.54	0	0	0.79	0.89	0.5	1.7	0.55	1.7	1.49	1.23	0.85	1.07	1.05
E2007	0	1.06	0.94	0.94	0.74	1.48	0	1.56	0.8	1.58	1.26	1.34	0	1.04	1.1	1.2	1.25	1.48
E1578	0	0	1.5	1.5	0	1.35	1.41	1.86	1.29	0.93	1.44	1.88	1.62	1.75	1.91	1.48	0.66	1.2

WHAT IS ASSIGNMENT PROBLEM?

ASSIGNMENT PROBLEM – FLOW DIAGRAM

a - assignees

t - tasks



ASSUMPTIONS

- Skill score is a fixed number.
- All tasks are equally importance.
- One worker can do at most 2 tasks.
- Additional of 2 workers.
- Workers are fixed for the given line.
- If the workers are less than 18, presented workers must be assigned for a task.

THE MATHEMATICAL MODEL

Case 01 – No. workers $\geq$ No. of tasks

Let x_{ij} be the number of units assign i^{th} worker to j^{th} task

Max $Z = \sum_{i=1}^{m} \sum_{j=1}^{n} S_{ij}\, x_{ij}$

Subject to

$\sum_{i=1}^{m} x_{ij} \leq 1$ for i = 1,2,………,n..,m

$\sum_{j=1}^{n} x_{ij} = 1$ for j = 1,2,………,n

$$x_{ij} = \begin{cases} 1 & ;\text{ if assigning } i^{th} \text{ worker to } j^{th} \text{ task} \\ 0 & ;\text{ if not assigning} \end{cases}$$

Case 02 : No. workers < No. of tasks

Let x_{ij} be the number of units assign i^{th} worker to j^{th} task

Max $Z = \sum_{i=1}^{m} \sum_{j=1}^{n} S_{ij}\, x_{ij}$

Subject to

$\sum_{i=1}^{m} x_{ij} \leq 2$ for i = 1,2,……….,n...,m

$\sum_{j=1}^{n} x_{ij} = 1$ for j = 1,2,……….,n

$\sum_{i=1}^{m}(\sum_{j=1}^{n} x_{ij} = 2)$ = No. absents - 2

$$x_{ij} = \begin{cases} 1 & \text{; if assigning } i^{th} \text{ worker to } j^{th} \text{ task} \\ 0 & \text{; if not assigning} \end{cases}$$

MODEL SOLUTION SOFTWARES

Excel

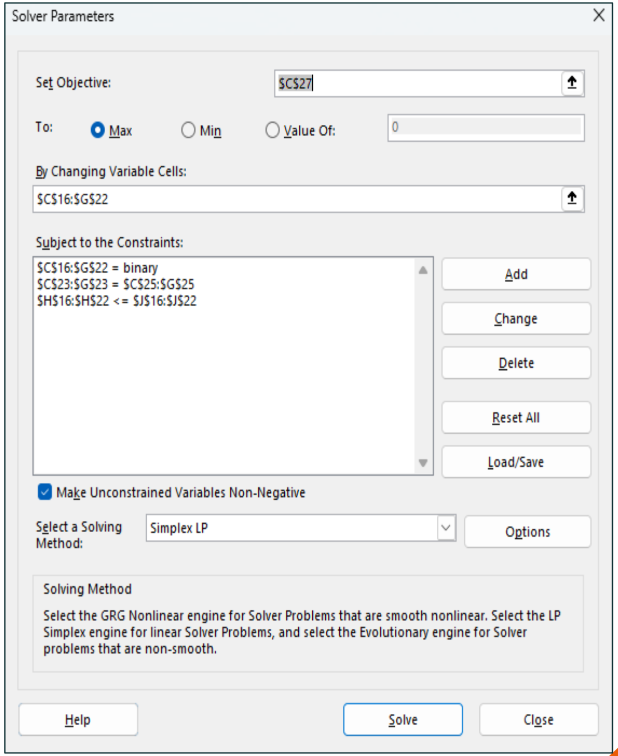
Python

SECONDARY DATASET

	Task 0	Task 1	Task 2	Task 3	Task 4
EMP001	1.25	0	1.65	0	0.66
EMP002	0.82	0	0	1.5	0.99
EMP003	0	1.98	0	1.45	0
EMP004	1.39	1.4	0	0.9	0
EMP005	1.3	0	0.69	0	0
EMP006	1.56	1.33	0	1.9	1.96
EMP007	0	0.79	1.4	0	0.86

VISUAL REPRESENTATION IN EXCEL

		Tasks							
		T0	T1	T2	T3	T4	Row Sum		
Employees	Emp1	0	0	1	0	0	1	<=	1
	Emp2	0	0	0	1	0	1	<=	1
	Emp3	0	1	0	0	0	1	<=	1
	Emp4	1	0	0	0	0	1	<=	1
	Emp5	0	0	0	0	0	0	<=	1
	Emp6	0	0	0	0	1	1	<=	1
	Emp7	0	0	0	0	0	0	<=	1
	Column Sum	1	1	1	1	1			
		=	=	=	=	=			
		1	1	1	1	1			
	Objective Function	8.48							



VISUAL REPRESENTATION IN PYTHON

```
Are there any absent employees? (yes/no): no

Skill Score Dataset (Worker ID and Skill Scores for each task):
  Worker ID  Task 0  Task 1  Task 2  Task 3  Task 4
0    EMP001    1.25    0.00    1.65    0.00    0.66
1    EMP002    0.82    0.00    0.00    1.50    0.99
2    EMP003    0.00    1.98    0.00    1.45    0.00
3    EMP004    1.39    1.40    0.00    0.90    0.00
4    EMP005    1.30    0.00    0.69    0.00    0.00
5    EMP006    1.56    1.33    0.00    1.90    1.96
6    EMP007    0.00    0.79    1.40    0.00    0.86

Worker Task Assignments and Skill Scores:
  Worker ID    Task  Skill Score
0    EMP001  Task 2         1.65
1    EMP002  Task 3         1.50
2    EMP003  Task 1         1.98
3    EMP004  Task 0         1.39
4    EMP006  Task 4         1.96

Total Maximized Skill Score: 8.48
```

PYTHON CODE IMPLEMENTATION

- Import packages

```
import openpyxl
import pandas as pd
import numpy as np
from scipy.optimize import linear_sum_assignment
from heapq import heappush, heappop
```

- Create Worker Class

```
class Worker:
    def __init__(self, id):
        self.id = id
        self.tasks = []

    def assign_task(self, task_index):
        self.tasks.append(task_index)

    def can_take_more_tasks(self):
        return len(self.tasks) < 2

    def __str__(self):
        return f"Worker {self.id}: Tasks {self.tasks}"
```

- Load Skill Scores from Excel File

```
def load_skill_scores_from_excel(file_path):
    workbook = openpyxl.load_workbook(file_path)
    sheet = workbook.active
    task_names = [cell.value for cell in sheet[1]][1:]
    workers_data = []
    for row in sheet.iter_rows(min_row=2, values_only=True):
        worker_id = row[0]
        skill_scores = list(row[1:])
        workers_data.append([worker_id] + skill_scores)
    df = pd.DataFrame(workers_data, columns=['Worker ID'] + task_names)
    return df
```

- Algorithm

```python
# More Workers or Equal Case
def hungarian_task_assignment(skill_matrix, num_tasks):

    num_workers = skill_matrix.shape[0]
    cost_matrix = skill_matrix * -1
    row_ind, col_ind = linear_sum_assignment(cost_matrix)

    assignments = [(row, col) for row, col in zip(row_ind, col_ind)]
    total_skill_score = skill_matrix[row_ind, col_ind].sum()

    return assignments, total_skill_score
```

- # Algorithm

```python
# Task Assignment with Constraints for fewer workers than tasks
def maximize_skill_score_with_constraints(skill_matrix, num_tasks, absent_count):
    num_workers = skill_matrix.shape[0]
    max_multiple_tasks_workers = absent_count -2 # Strictly less than the number of absent employees
    workers_with_multiple_tasks = 0

    task_heap = []
    for i in range(num_workers):
        for j in range(num_tasks):
            if skill_matrix[i, j] > 0:
                heappush(task_heap, (-skill_matrix[i, j], i, j))

    assignments = []
    total_skill_score = 0
    workers = [Worker(i) for i in range(num_workers)]
    assigned_tasks = set()

    while len(assigned_tasks) < num_tasks:
        if task_heap:
            neg_score, worker_id, task_id = heappop(task_heap)
            score = -neg_score

            if task_id in assigned_tasks or score == 0:
                continue
```

Ctd…

- Algorithm

```
        max_tasks = 2 if (len(workers[worker_id].tasks) == 0 and workers_with_multiple_tasks < max_multiple_tasks_workers) else 1
        if workers[worker_id].can_take_more_tasks(max_tasks):

            workers[worker_id].assign_task(task_id, score)
            assigned_tasks.add(task_id)
            total_skill_score += score
            assignments.append((worker_id, task_id))


            if len(workers[worker_id].tasks) == 2:
                workers_with_multiple_tasks += 1
    else:

        for worker_id in range(num_workers):
            for task_id in range(num_tasks):
                if task_id not in assigned_tasks and skill_matrix[worker_id, task_id] > 0:
                    workers[worker_id].assign_task(task_id, skill_matrix[worker_id, task_id])
                    assigned_tasks.add(task_id)
                    total_skill_score += skill_matrix[worker_id, task_id]
                    assignments.append((worker_id, task_id))
                    break

return assignments, total_skill_score
```

SOLUTIONS

CASE 01 – NO. WORKERS ≥ NO. OF TASKS

- Case A – No Absences

```
Are there any absent employees? (yes/no): no

Worker Task Assignments and Skill Scores:

   Worker ID                                  Task  Skill Score
0      E1118                   sewing front pieces         1.85
1      E1246                               ironing         1.95
2      E1156                            packing(1)         1.95
3      E1143                  Sewing back pieces(1)        1.58
4      E1179                            Sewing hem         1.89
5      E1099  Finishing front pieces & Thread cutting      1.87
6      E1233                           sewing pouch        1.85
7      E0999                           Attach pouch        1.95
8      E1567                        closing elastic        1.94
9      E1589              binding fold over elastic        1.86
10     E2004                  Sewing back pieces(2)        1.84
11     E2014                            packing(2)         1.94
12     E1887                   sewing front pieces         1.77
13     E1934                        Sewing elastic         1.50
14     E1469    Finishing back piece & Thread cutting      1.91
15     E2069                    cutting elastic band       1.94
16     E2071                     Front pieces darts        1.94
17     E1578                        Finishing pouch        1.86

Total Maximized Skill Score: 33.39
```

Optimal Skill Score

Case B – 1 worker absent

```
Are there any absent employees? (yes/no): yes
Enter the IDs of absent employees (comma-separated): E1118

Worker Task Assignments and Skill Scores:

   Worker ID                                     Task  Skill Score
0      E1246                               packing(2)         1.75
1      E1156                               packing(1)         1.95
2      E1143                     Sewing back pieces(1)        1.58
3      E1179                               Sewing hem         1.89
4      E1099  Finishing front pieces & Thread cutting         1.87
5      E1233                             sewing pouch         1.85
6      E0999                             Attach pouch         1.95
7      E1567                          closing elastic         1.94
8      E1589                 binding fold over elastic         1.86
9      E2004                     Sewing back pieces(2)        1.84
10     E2014                       sewing front pieces        1.80
11     E1887                       sewing front pieces        1.77
12     E1934                                  ironing         1.88
13     E1469    Finishing back piece & Thread cutting         1.91
14     E2069                      cutting elastic band        1.94
15     E2071                        Front pieces darts        1.94
16     E2007                           Finishing pouch        1.56
17     E1578                            Sewing elastic        1.75

Total Maximized Skill Score: 33.03
```

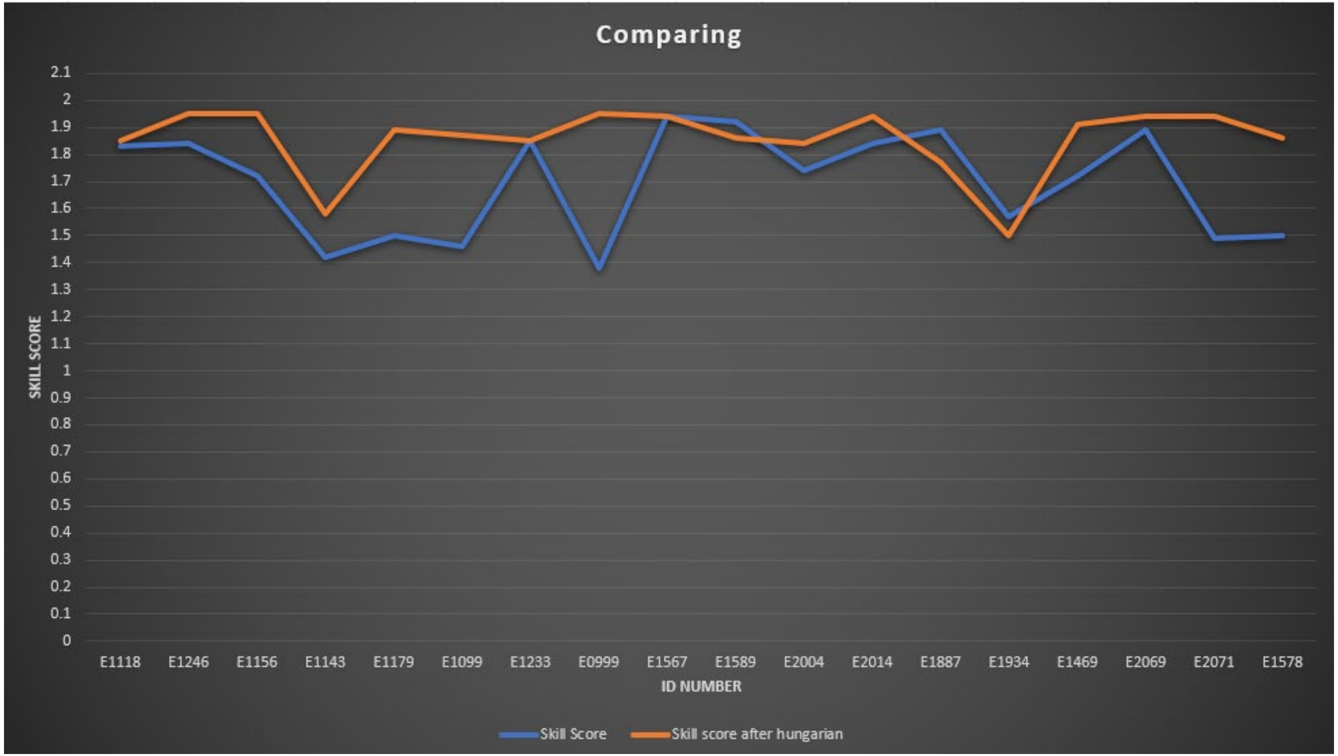
CASE 02 – NO. WORKERS < NO. OF TASKS

```
Are there any absent employees? (yes/no): yes
Enter the absent employee IDs (comma-separated): E0999,E1567,E2071,E1246

Worker Task Assignments and Skill Scores:
   Worker ID                                    Task  Skill Score
0      E1156                              packing(1)         1.95
1      E2014                              packing(2)         1.94
2      E2069                     cutting elastic band         1.94
3      E1589                    Sewing back pieces(2)         1.92
4      E1887                       Front pieces darts         1.91
5      E1469    Finishing back piece & Thread cutting         1.91
6      E1578                               Sewing hem         1.91
7      E1934                                  ironing         1.88
8      E1099  Finishing front pieces & Thread cutting         1.87
9      E1179                binding fold over elastic         1.86
10     E1118                      sewing front pieces         1.85
11     E1233                             sewing pouch         1.85
12     E1143                    Sewing back pieces(1)         1.58
13     E2007                          Finishing pouch         1.56
14     E1857                          closing elastic         1.51
15     E2004                             Attach pouch         0.92
16     E1118                      sewing front pieces         1.85
17     E1156                           Sewing elastic         1.14

Total Maximized Skill Score: 31.350000000000001
```

COMPARISON

EMPLOEE ID	COMPANY'S ASSIGNMENT	OUR ASSIGNMENT
E1118	1.83	1.85
E1246	1.84	1.95
E1156	1.72	1.95
E1143	1.42	1.58
E1179	1.5	1.89
E1099	1.46	1.87
E1233	1.85	1.85
E0999	1.38	1.95
E1567	1.94	1.94
E1589	1.92	1.86
E2004	1.74	1.84
E2014	1.84	1.94
E1887	1.89	1.77
E1934	1.57	1.5
E1469	1.72	1.91
E2069	1.89	1.94
E2071	1.49	1.94
E1578	1.5	1.86
Total Skill Score	**30.5**	**33.39**

GRAPHICAL REPRESENTATION



BENEFITS OF THE MODEL

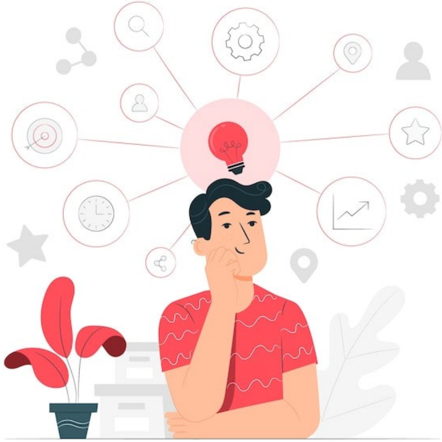
- **For the Factory:**
  - Enhanced productivity.
  - Better use of skilled labor.
  - Reduces production time.
  - Reduces production costs.
- **For Workers:**
  - Tasks aligned with skills.
  - Fair workload distribution.

DIFFICULTIES FACED

- Data collection challenges.
- Complex constraints.
- Model design.
- Computational complexity.
- Difficult to implement in Excel.

CONCLUSION

- Developed a model for optimal task assignment of workers.
- Reduced the effect of workers absences.
- Improved efficiency and productivity at Panvila Emjay Garment Factory.



TIMELINE

Task	Month									
	April	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan
Finding a place to continue the project	■									
Visited Emjay international pvt(ltd)		■								
Selected the project topic and identified the problem		■	■							
Meeting 1			■							
Data preparation			■	■						
Online meeting				■						
Analyse the skill scores and identify trends or patterns.					■					
Design the model using hungarian method					■					
Meeting 2						■				
Mid presentation						■				
Implement the algorithom in python						■	■			
Test the model with using sample data in excel and python							■			
Validate the results with actual factory scenarioes							■	■	■	
Prepare the final presentation and final report									■	■
Meeting 3										■
Final presentation										■

REFERENCES

Https://en.Wikipedia.Org/wiki/hungarian_algorithm

Https://www.Srilankabusiness.Com/exporters-directory/company-profiles/emjay-international-pvt-ltd/

Https://www.Srilankabusiness.Com/apparel/

Https://kanchiuniv.Ac.In/coursematerials/operationresearch.Pdf

THANK YOU